ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 4 cm	$PT^2 = OP^2 - OT^2 = 5^2 - 3^2 = 16$; $PT = 4$ cm
2	**(c)** 6 cm	$r^2 = OP^2 - PT^2 = 10^2 - 8^2 = 36$; $r = 6$ cm
3	**(b)** 25 cm	$OP^2 = r^2 + PT^2 = 7^2 + 24^2 = 625$; $OP = 25$ cm
4	**(a)** 14 cm	Perimeter $= PA + AO + OB + BP = 2(4) + 2(3)$; Perimeter $= 14$ cm
5	**(b)** 70°	In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 110^\circ$; $\angle APB = 70^\circ$
6	**(c)** 60°	$\angle AOP = 180^\circ - 90^\circ - 30^\circ$; $\angle AOP = 60^\circ$
7	**(c)** one point	Definition of a tangent
8	**(b)** 8 cm	$PT^2 = OP^2 - OT^2 = 10^2 - 6^2 = 64$; $PT = 8$ cm
9	**(d)** 5 cm	$r^2 = OP^2 - PT^2 = 13^2 - 12^2 = 25$; $r = 5$ cm
10	**(c)** 25 cm	$OP^2 = r^2 + PT^2 = 15^2 + 20^2 = 625$; $OP = 25$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 25^2 - 20^2 = 225$ (1); $PT = 15$ cm (1).	1+1
12	Radius $\perp$ tangent, so OP is the hypotenuse of a right triangle with sides r and PT; $OP^2 = r^2 + PT^2 = 4^2 + 3^2 = 25$ (1); $OP = 5$ cm (1).	1+1
13	$OA \perp PA$ and $OB \perp PB$, so $\angle OAP = \angle OBP = 90^\circ$; In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 120^\circ$ (1); $\angle APB = 60^\circ$ (1).	1+1
14	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 26^2 - 24^2 = 100$ (1); $PT = 10$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter $= PA + AO + OB + BP = 2(16) + 2(12)$ (1); Perimeter $= 56$ cm (1).	1+1+1
16	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 3 \times 4$ (1); Area $= 12$ cm^2 (1).	1+1+1
17	$OA \perp PA$, so $\triangle OAP$ is right-angled at A (1); $\angle AOP = 180^\circ - 90^\circ - 20^\circ$ (1); $\angle AOP = 70^\circ$ (1).	1+1+1
	OR: $PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter $= PA + AO + OB + BP = 2(24) + 2(7)$ (1); Perimeter $= 62$ cm (1).	1+1+1
18	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 6 \times 8$ (1); Area $= 48$ cm^2 (1).	1+1+1
	OR: $OA \perp PA$, so $\triangle OAP$ is right-angled at A (1); $\angle AOP = 180^\circ - 90^\circ - 45^\circ$ (1); $\angle AOP = 45^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{5^2 - 3^2} = 4$ cm; perimeter $= 2(4 + 3)$ (2); Perimeter 14 cm; area $= 2 \times \frac{1}{2} \times 3 \times 4 = 12$ cm^2 (1).	2+2+1
	OR: The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (2); Half-chord $= \sqrt{25^2 - 7^2} = 24$ cm (2); Chord $= 2 \times 24 = 48$ cm (1).	2+2+1
20	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{10^2 - 6^2} = 8$ cm; perimeter $= 2(8 + 6)$ (2); Perimeter 28 cm; area $= 2 \times \frac{1}{2} \times 6 \times 8 = 48$ cm^2 (1).	2+2+1
	OR: The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (2); Half-chord $= \sqrt{25^2 - 15^2} = 20$ cm (2); Chord $= 2 \times 20 = 40$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90^\circ$; the correct sum is 180°.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 28 cm	Perimeter $= PA + AO + OB + BP = 2(8) + 2(6)$; Perimeter $= 28$ cm
2	**(d)** 80°	In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 100^\circ$; $\angle APB = 80^\circ$
3	**(a)** 55°	$\angle AOP = 180^\circ - 90^\circ - 35^\circ$; $\angle AOP = 55^\circ$
4	**(b)** 90°	Radius $\perp$ tangent
5	**(c)** 12 cm	$PT^2 = OP^2 - OT^2 = 13^2 - 5^2 = 144$; $PT = 12$ cm
6	**(a)** 9 cm	$r^2 = OP^2 - PT^2 = 15^2 - 12^2 = 81$; $r = 9$ cm
7	**(c)** 26 cm	$OP^2 = r^2 + PT^2 = 10^2 + 24^2 = 676$; $OP = 26$ cm
8	**(c)** 34 cm	Perimeter $= PA + AO + OB + BP = 2(12) + 2(5)$; Perimeter $= 34$ cm
9	**(d)** 50°	In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 130^\circ$; $\angle APB = 50^\circ$
10	**(c)** 50°	$\angle AOP = 180^\circ - 90^\circ - 40^\circ$; $\angle AOP = 50^\circ$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Radius $\perp$ tangent, so OP is the hypotenuse of a right triangle with sides r and PT; $OP^2 = r^2 + PT^2 = 8^2 + 6^2 = 100$ (1); $OP = 10$ cm (1).	1+1
12	$OA \perp PA$ and $OB \perp PB$, so $\angle OAP = \angle OBP = 90^\circ$; In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 90^\circ$ (1); $\angle APB = 90^\circ$ (1).	1+1
13	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 29^2 - 21^2 = 400$ (1); $PT = 20$ cm (1).	1+1
14	Radius $\perp$ tangent, so OP is the hypotenuse of a right triangle with sides r and PT; $OP^2 = r^2 + PT^2 = 12^2 + 5^2 = 169$ (1); $OP = 13$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter $= PA + AO + OB + BP = 2(20) + 2(15)$ (1); Perimeter $= 70$ cm (1).	1+1+1
16	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 5 \times 12$ (1); Area $= 60$ cm^2 (1).	1+1+1
17	$OA \perp PA$, so $\triangle OAP$ is right-angled at A (1); $\angle AOP = 180^\circ - 90^\circ - 50^\circ$ (1); $\angle AOP = 40^\circ$ (1).	1+1+1
	OR: $PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter $= PA + AO + OB + BP = 2(24) + 2(10)$ (1); Perimeter $= 68$ cm (1).	1+1+1
18	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 9 \times 12$ (1); Area $= 108$ cm^2 (1).	1+1+1
	OR: $OA \perp PA$, so $\triangle OAP$ is right-angled at A (1); $\angle AOP = 180^\circ - 90^\circ - 60^\circ$ (1); $\angle AOP = 30^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{13^2 - 5^2} = 12$ cm; perimeter $= 2(12 + 5)$ (2); Perimeter 34 cm; area $= 2 \times \frac{1}{2} \times 5 \times 12 = 60$ cm^2 (1).	2+2+1
	OR: The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (2); Half-chord $= \sqrt{26^2 - 10^2} = 24$ cm (2); Chord $= 2 \times 24 = 48$ cm (1).	2+2+1
20	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{15^2 - 9^2} = 12$ cm; perimeter $= 2(12 + 9)$ (2); Perimeter 42 cm; area $= 2 \times \frac{1}{2} \times 9 \times 12 = 108$ cm^2 (1).	2+2+1
	OR: The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (2); Half-chord $= \sqrt{29^2 - 20^2} = 21$ cm (2); Chord $= 2 \times 21 = 42$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90°$; the correct sum is $180°$.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 0	Every line through an inside point cuts the circle in two points
2	**(b)** 12 cm	$PT^2 = OP^2 - OT^2 = 15^2 - 9^2 = 144$; $PT = 12$ cm
3	**(d)** 8 cm	$r^2 = OP^2 - PT^2 = 17^2 - 15^2 = 64$; $r = 8$ cm
4	**(d)** 29 cm	$OP^2 = r^2 + PT^2 = 20^2 + 21^2 = 841$; $OP = 29$ cm
5	**(c)** 42 cm	Perimeter $= PA + AO + OB + BP = 2(12) + 2(9)$; Perimeter $= 42$ cm
6	**(a)** 40°	In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 140^\circ$; $\angle APB = 40^\circ$
7	**(c)** 35°	$\angle AOP = 180^\circ - 90^\circ - 55^\circ$; $\angle AOP = 35^\circ$
8	**(a)** 1	Only one tangent at a point of the circle
9	**(b)** 15 cm	$PT^2 = OP^2 - OT^2 = 17^2 - 8^2 = 225$; $PT = 15$ cm
10	**(a)** 12 cm	$r^2 = OP^2 - PT^2 = 20^2 - 16^2 = 144$; $r = 12$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$OA \perp PA$ and $OB \perp PB$, so $\angle OAP = \angle OBP = 90^\circ$; In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 150^\circ$ (1); $\angle APB = 30^\circ$ (1).	1+1
12	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 30^2 - 24^2 = 324$ (1); $PT = 18$ cm (1).	1+1
13	Radius $\perp$ tangent, so OP is the hypotenuse of a right triangle with sides r and PT; $OP^2 = r^2 + PT^2 = 12^2 + 9^2 = 225$ (1); $OP = 15$ cm (1).	1+1
14	$OA \perp PA$ and $OB \perp PB$, so $\angle OAP = \angle OBP = 90^\circ$; In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 80^\circ$ (1); $\angle APB = 100^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter $= PA + AO + OB + BP = 2(21) + 2(20)$ (1); Perimeter $= 82$ cm (1).	1+1+1
16	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 8 \times 15$ (1); Area $= 120$ cm^2 (1).	1+1+1
17	$OA \perp PA$, so $\triangle OAP$ is right-angled at A (1); $\angle AOP = 180^\circ - 90^\circ - 15^\circ$ (1); $\angle AOP = 75^\circ$ (1).	1+1+1
	OR: $PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter $= PA + AO + OB + BP = 2(24) + 2(18)$ (1); Perimeter $= 84$ cm (1).	1+1+1
18	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 12 \times 16$ (1); Area $= 192$ cm^2 (1).	1+1+1
	OR: $OA \perp PA$, so $\triangle OAP$ is right-angled at A (1); $\angle AOP = 180^\circ - 90^\circ - 65^\circ$ (1); $\angle AOP = 25^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{17^2 - 8^2} = 15$ cm; perimeter $= 2(15 + 8)$ (2); Perimeter 46 cm; area $= 2 \times \frac{1}{2} \times 8 \times 15 = 120$ cm^2 (1).	2+2+1
	OR: The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (2); Half-chord $= \sqrt{30^2 - 18^2} = 24$ cm (2); Chord $= 2 \times 24 = 48$ cm (1).	2+2+1
20	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{20^2 - 12^2} = 16$ cm; perimeter $= 2(16 + 12)$ (2); Perimeter 56 cm; area $= 2 \times \frac{1}{2} \times 12 \times 16 = 192$ cm^2 (1).	2+2+1
	OR: The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (2); Half-chord $= \sqrt{10^2 - 8^2} = 6$ cm (2); Chord $= 2 \times 6 = 12$ cm (1).	2+2+1

Mistakes to avoid:

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- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90°$; the correct sum is $180°$.
- Assuming a tangent can be drawn from a point inside the circle.

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